

Day 1 — The Sūtra — All from 9, Last from 10

Module: Subtraction — Nikhilaṃ (All from 9) · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: compute the complement of any 2-digit number to 100.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilaṃ Navataścaramaṃ Daśataḥ* (IAST: Nikhilaṃ Navataścaramaṃ Daśataḥ; lit. “all from 9, the last from 10”)

Primary source (today’s reading)

From *Bharati Krishna Tirthaji* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “What’s one thing you remember about yesterday’s lesson?”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Pose the question.** “What’s $100 - 47$?” Most students compute 53 by borrowing. Ask: “What’s $1000 - 437$?” Trickier — the borrowing chains. Today’s method removes the chain entirely.
2. **State the sūtra.** *Nikhilaṃ Navataścaramaṃ Daśataḥ* — “all from 9, the last from 10.” Write the Sanskrit and IAST on the board.
3. **Apply to $100 - 47$.** Subtract each digit of 47 from 9, except the last which is from 10:
 - 4 from 9 = 5
 - 7 from 10 = 3
 - Answer: **53**. No borrowing.
4. **Pattern recognition.** For $1000 - 437$: $9-4=5$, $9-3=6$, $10-7=3 \rightarrow$ **563**. Verify: $437 + 563 = 1000$. ✓

5. **Source note.** Tirthaji's *Vedic Mathematics* (1965), sūtra #2. Honest history: this is one of the more solidly-rooted methods in older Indian and Arabic arithmetic traditions (decimal complement was widely used pre-modern). Independent practice: 10 problems on the worksheet.

Activity (15 min)

A scaled version of *Complement Flashcards* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces The Sūtra — All from 9, Last from 10.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).